41. Robust Bash: errexit, nounset, traps

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Phase: Phase 5 — Shell scripting
Estimated time: 1–2 days
Slug: `41-bash-robust`

Learning objectives

- Use traps for cleanup
- Handle signals in scripts
- Create temp dirs safely
- Write scripts that fail loudly

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

```
set -euo pipefail
tmp=$(mktemp -d)
trap 'rm -rf "$tmp"' EXIT

work() { ...; }
work
```

traps ensure cleanup on exit/signals. Combined with nounset, many silent bugs become loud.

Common mistakes

- rm -rf unprotected tmp
- Ignoring ERR traps for critical scripts
- set +e hiding failures permanently

Exercises

Work in `exercises/41-bash-robust/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/41-bash-robust/` until you have an honest failing attempt.

Self-check

1. What does trap EXIT do?

2. mktemp why?
3. nounset catches what?

Further reading

- https://mywiki.wooledge.org/SignalTrap